

What Is Natural Language Processing (NLP)?

Part I - Introduction

Topic: What Is Natural Language Processing (NLP)?

The course presents NLP as a field concerned with interactions between computers and human language.

It focuses on programming computers to process and analyze large amounts of natural-language data.

What Is Natural Language Processing?

Natural Language Processing (NLP) is a subfield of linguistics, computer science, and artificial intelligence.

It is concerned with the interactions between computers and human language.

In particular, NLP addresses how to program computers to process and analyze large amounts of natural-language data.

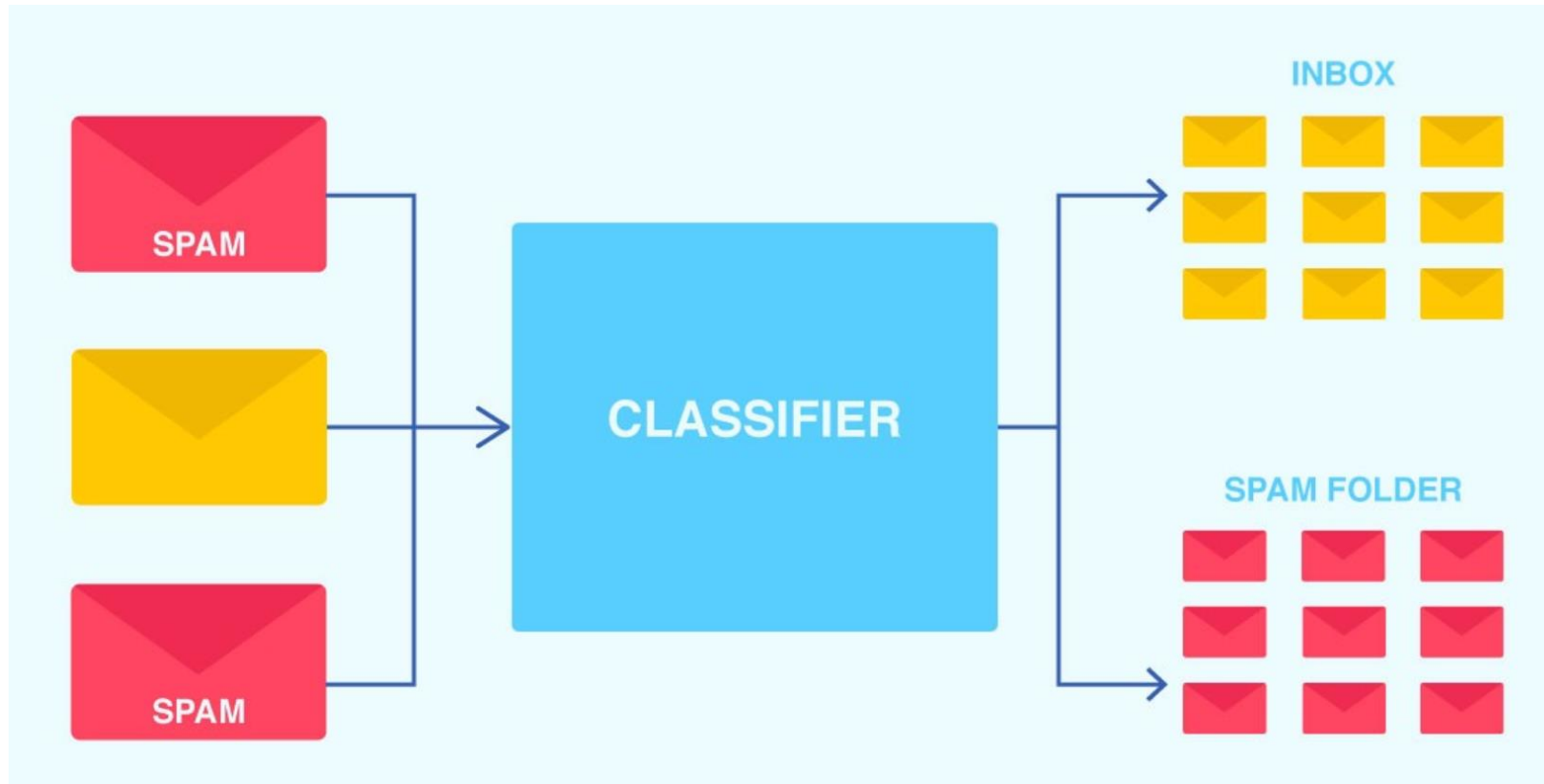
NLP - Scope

NLP is presented as an interdisciplinary field combining:

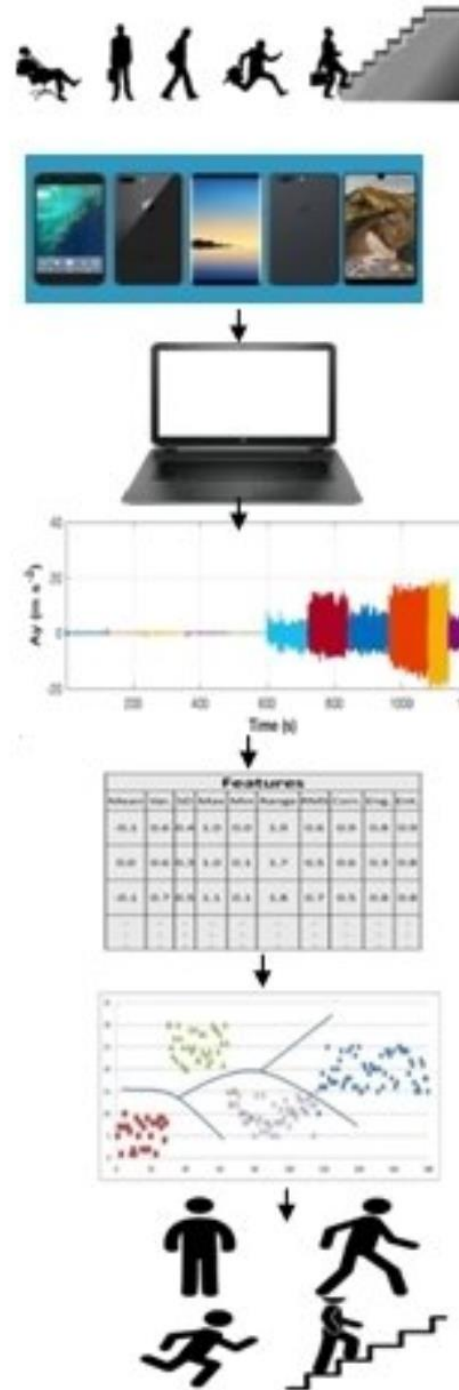
- Linguistics
- Computer Science
- Artificial Intelligence

The central objective is to enable computers to interact with, process, and analyze human language.

Example Application : Spam Classification



Example of a Data Processing Workflow



Why Is NLP Important?

How important is NLP in mobile?

Statistics courtesy of
Kleiner Perkins' Internet
Trends 2017

mobile
queries
made by
voice in
2016

20%

70%
of queries made
using conversational
language

95%

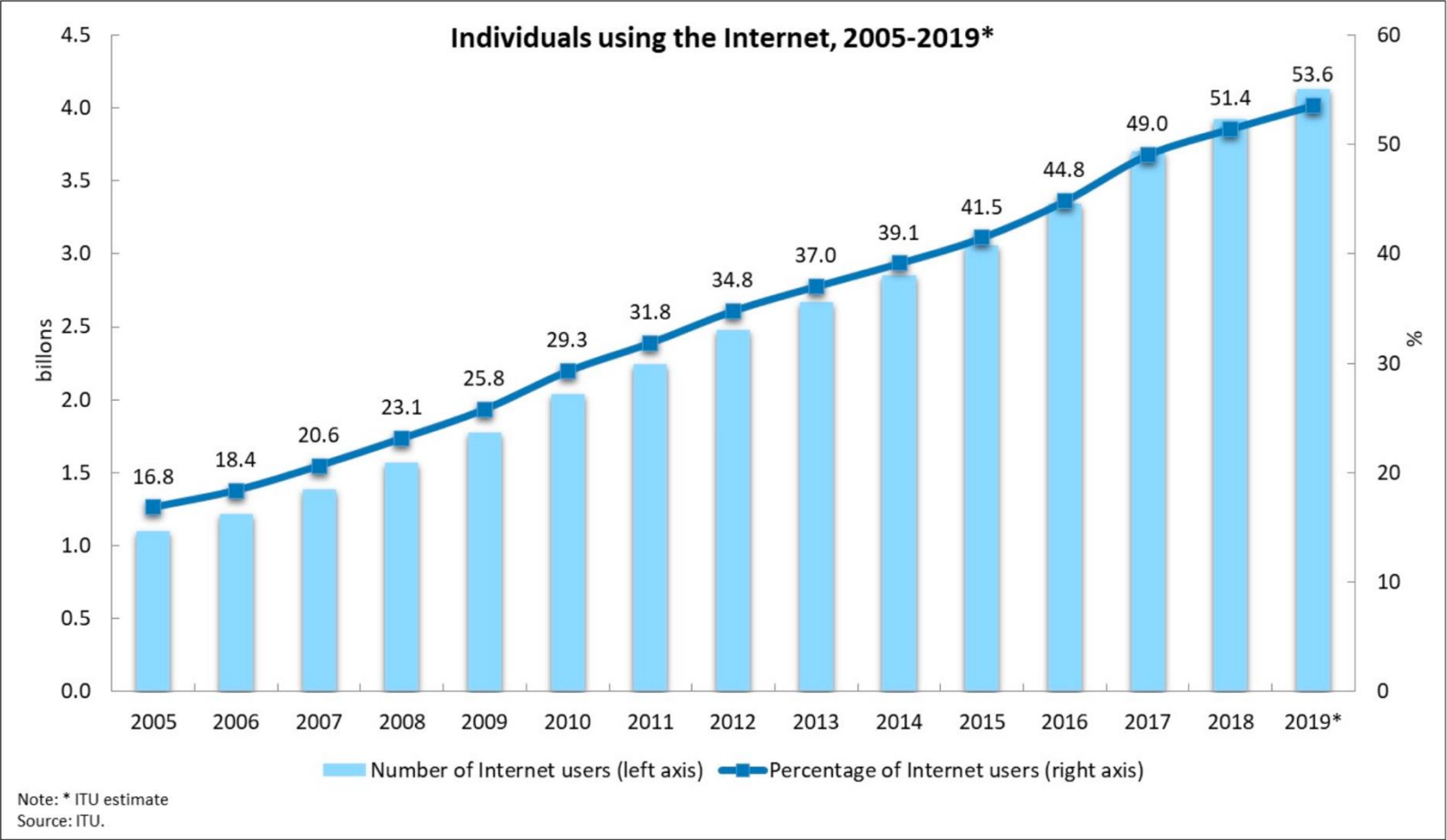
Accuracy required
for NLP tech to
meet human
tolerance levels

82%

would stop using a brand
after a bad experience

telecoms.com

Global Growth in Internet Use



Main Conclusions

NLP connects linguistics, computer science, and artificial intelligence.

Its focus is the interaction between computers and human language.

A central objective is processing and analyzing large amounts of natural-language data.

In this section, we illustrated contextual applications through classification, data-processing workflows, mobile NLP, and Internet-use statistics.

Key Terminology

Natural Language Processing (NLP)

Linguistics

Computer Science

Artificial Intelligence

Human Language

Natural-Language Data

Text Processing

Text Analysis

Classification

A Brief History of Natural Language Processing

1. Historical Roots of NLP

Natural Language Processing has historical roots that predate modern computing.

Philosophers such as Gottfried Wilhelm Leibniz and René Descartes proposed symbolic systems intended to establish correspondences between words across languages.

These proposals remained theoretical and did not lead to the construction of a practical machine.

Leibniz and Descartes — Early Symbolic Ideas



2. Early Ideas on Machine Translation

From the 1930s onward, various ideas and designs for machine translation systems began to emerge.

However, none of these early designs achieved genuinely effective practical performance.



1950: Alan Turing and the Turing Test

In 1950, Alan Turing published his influential paper, “Computing Machinery and Intelligence.”

The paper introduced what is now known as the Turing Test, proposed as a criterion related to machine intelligence.

The criterion evaluates whether a computer program can convincingly participate in a written conversation while imitating a human.

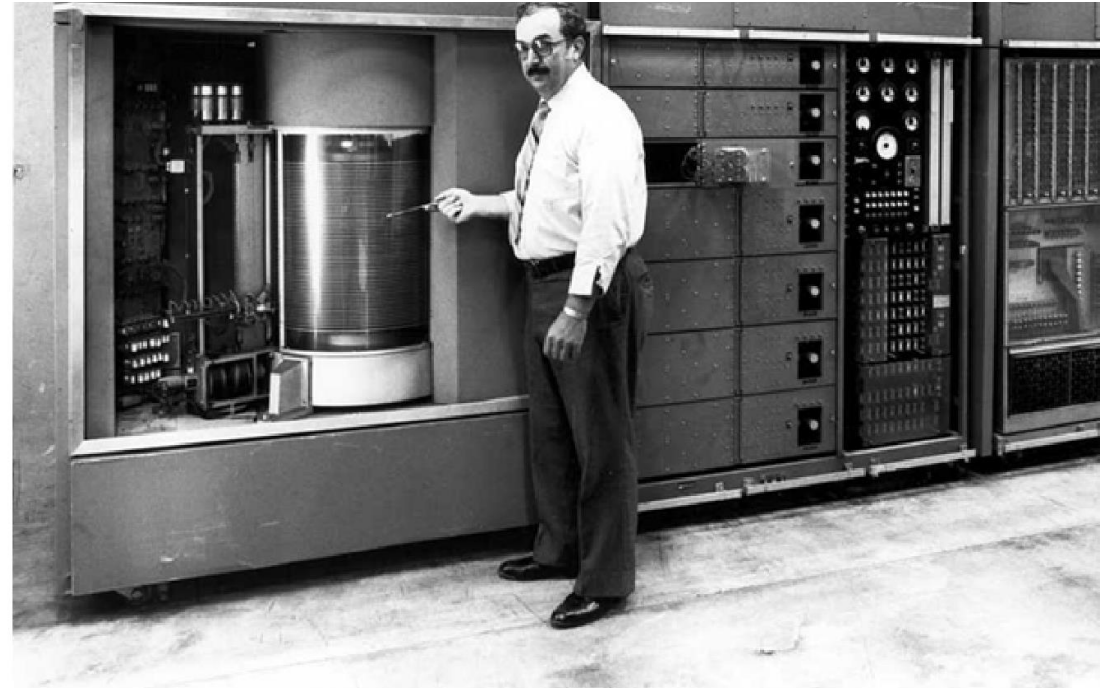
A human judge should be unable to reliably distinguish the program from a real human solely on the basis of the conversation.



1954 : The Georgetown–IBM Experiment

In 1954, the Georgetown–IBM experiment demonstrated a fully automatic translation system for more than sixty Russian sentences translated into English.

The authors of the program reportedly predicted that the machine-translation problem could be solved within three to five years.



1968 : SHRDLU

In 1968, Terry Winograd developed SHRDLU at MIT.

The system demonstrated a form of natural-language interaction with a computer.

The example shows the system interpreting commands and questions in a simulated blocks world.

The dialogue illustrates an early attempt at language understanding, reference resolution, and interaction with a structured environment.

Person: Grasp the pyramid.

Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.

Person (changing their mind): Find a block which is taller than the one you are holding and put it into the box.

Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.

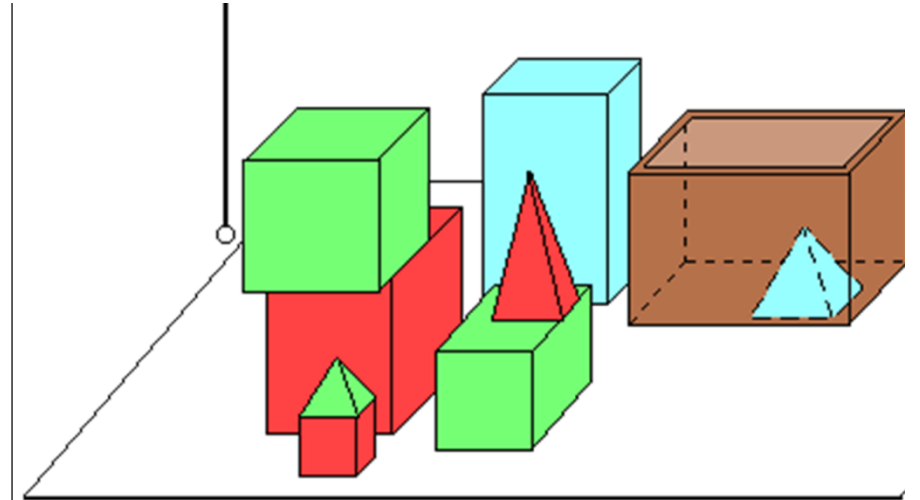
Computer: OK.

Person: What does the box contain?

Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.

Person: What is the pyramid supported by?

Computer: THE BOX.

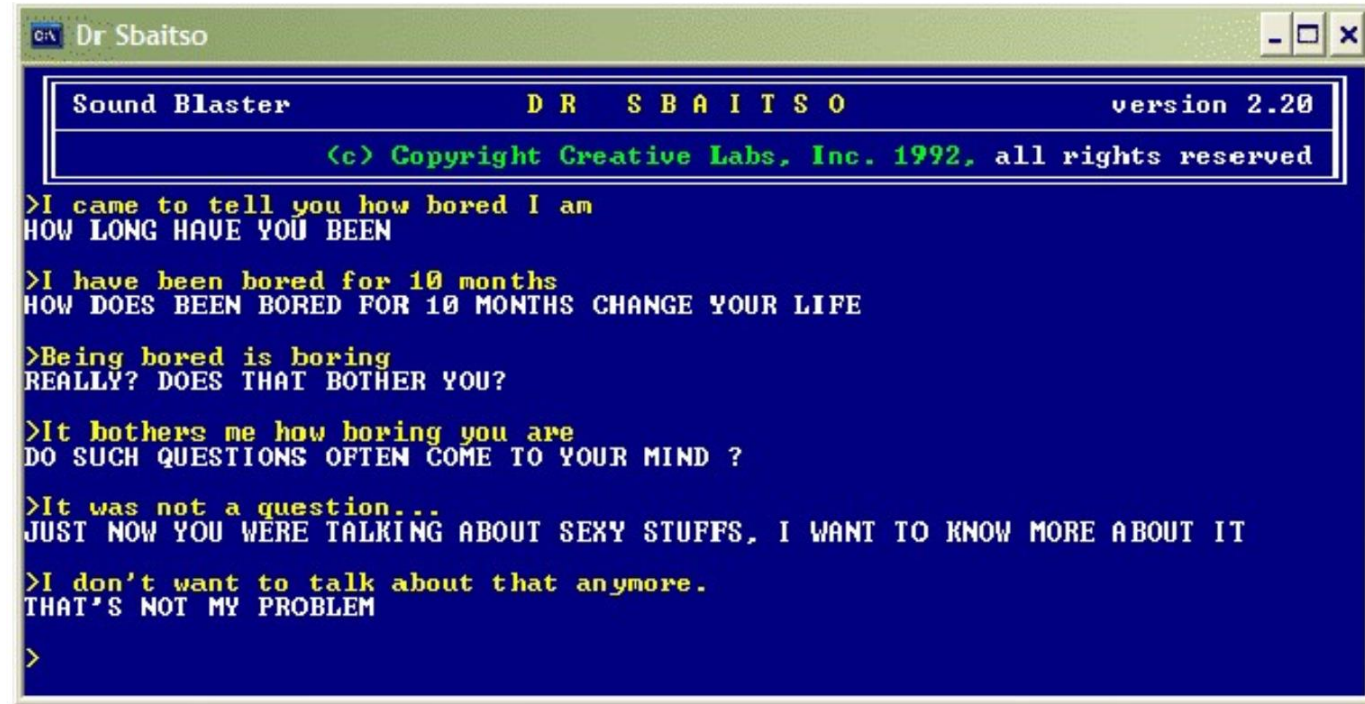


1991 : Dr. Sbaitso

In 1991, a prototype of an “intelligent psychotherapist” based on the chatbot concept was developed for DOS.

The system introduced itself as Dr. Sbaitso and invited the user to describe their thoughts and problems.

The example illustrates the evolution of conversational computer systems and early chatbot-style interaction.

A screenshot of a DOS window titled "Dr Sbaitso". The window has a green title bar and standard DOS window controls (minimize, maximize, close). The main area has a dark blue background with white and yellow text. At the top, there is a header bar with "Sound Blaster" on the left, "D R S B A I T S O" in the center, and "version 2.20" on the right. Below this is a line of green text: "<c> Copyright Creative Labs, Inc. 1992, all rights reserved". The chat history shows a series of user inputs in yellow and the chatbot's responses in white. The user inputs are: ">I came to tell you how bored I am", ">I have been bored for 10 months", ">Being bored is boring", ">It bothers me how boring you are", and ">It was not a question...". The chatbot's responses are: "HOW LONG HAVE YOU BEEN", "HOW DOES BEEN BORED FOR 10 MONTHS CHANGE YOUR LIFE", "REALLY? DOES THAT BOTHER YOU?", "DO SUCH QUESTIONS OFTEN COME TO YOUR MIND ?", and "JUST NOW YOU WERE TALKING ABOUT SEXY STUFFS, I WANT TO KNOW MORE ABOUT IT". The final user input is ">I don't want to talk about that anymore." and the chatbot's response is "THAT'S NOT MY PROBLEM". The prompt ">" is visible at the bottom left of the window.

```
Dr Sbaitso
Sound Blaster          D R S B A I T S O          version 2.20
(c) Copyright Creative Labs, Inc. 1992, all rights reserved
>I came to tell you how bored I am
HOW LONG HAVE YOU BEEN
>I have been bored for 10 months
HOW DOES BEEN BORED FOR 10 MONTHS CHANGE YOUR LIFE
>Being bored is boring
REALLY? DOES THAT BOTHER YOU?
>It bothers me how boring you are
DO SUCH QUESTIONS OFTEN COME TO YOUR MIND ?
>It was not a question...
JUST NOW YOU WERE TALKING ABOUT SEXY STUFFS, I WANT TO KNOW MORE ABOUT IT
>I don't want to talk about that anymore.
THAT'S NOT MY PROBLEM
>
```

2006 : IBM Watson

In 2006, IBM released the first version of Watson, its large-scale question-answering system.

Watson subsequently achieved a notable performance in answering questions on the American television quiz show Jeopardy!



2011-2017: The Rise of Personal Assistants

Siri was launched in 2011.

Amazon Alexa and Microsoft Cortana followed in 2014.

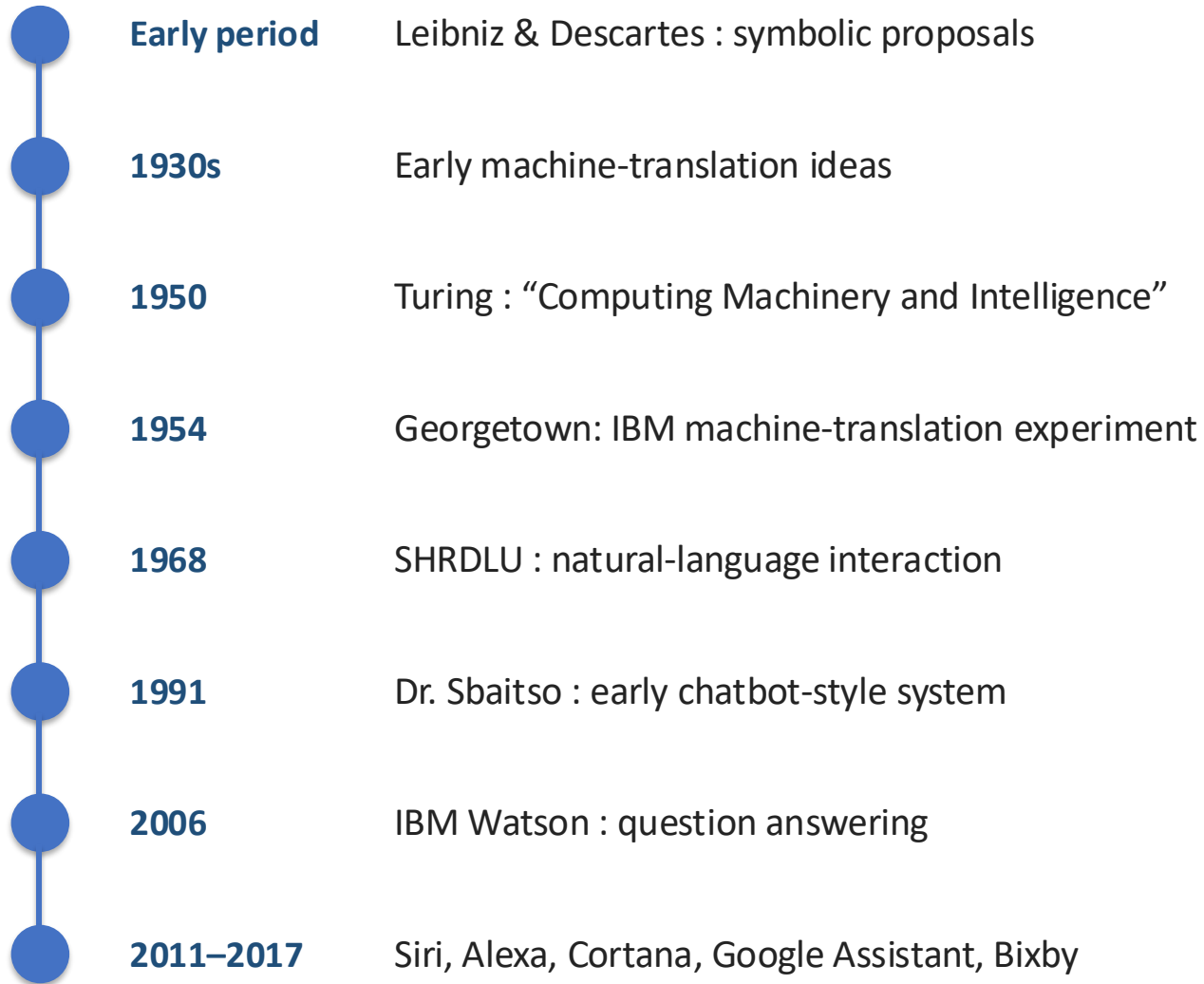
Google Assistant was introduced in 2016.

Samsung Bixby followed in 2017.

These systems represent the growing integration of natural-language interaction into consumer technologies.



NLP Historical Timeline



Main Conclusions

The history presented in this chapter shows a progression from theoretical ideas about symbolic language representation to machine translation, conversational systems, question answering, and voice-based personal assistants.

The examples illustrate successive attempts to make computers process, interpret, and respond to human language.

The chronology and descriptions in this presentation follow the provided course material.

Challenges in Natural Language Processing

Why Is Natural Language Understanding Difficult?

Natural Language Processing is one of the important and challenging areas of Deep Learning because it deals with language, which can be misunderstood even by humans.

Human language is not interpreted solely from individual words; meaning depends on structure, context, ambiguity, and shared knowledge.

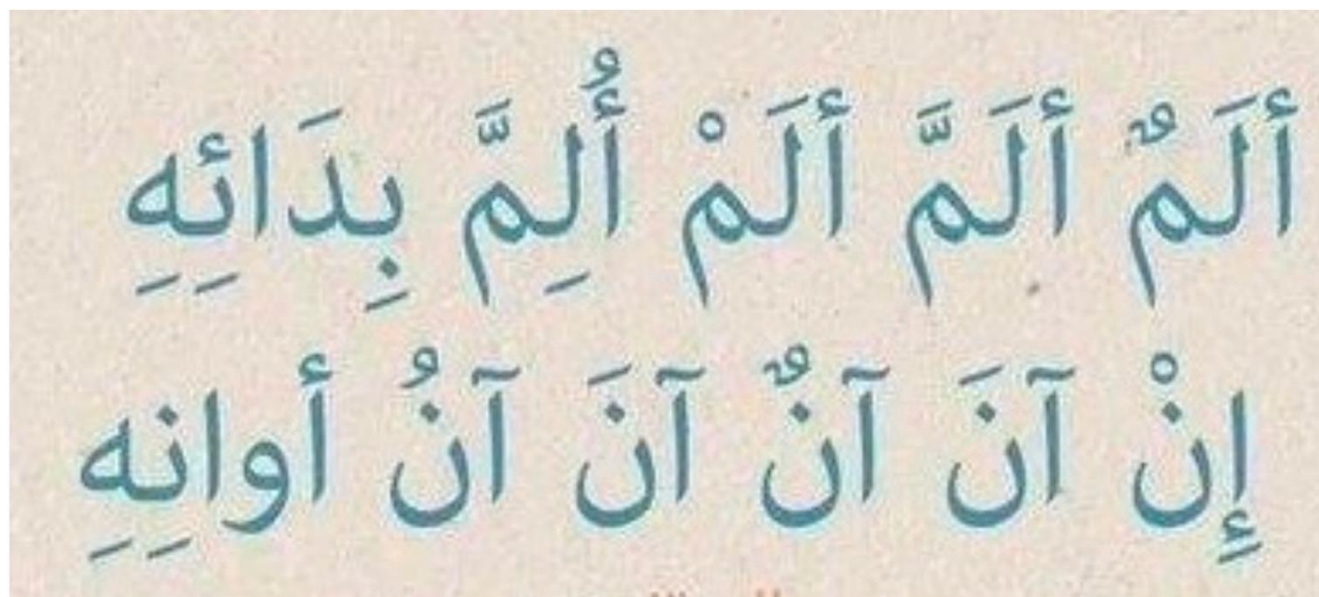
Computers handle numerical data and mathematical operations relatively easily, whereas natural-language text is unstructured and appears in many forms, domains, and languages.

Structural and Semantic Ambiguity

A sentence may be grammatically valid while still admitting multiple interpretations.

Ambiguity may arise from word boundaries, syntactic structure, lexical meaning, or the surrounding context.

This makes it difficult for an NLP system to determine the intended interpretation from the surface form alone.



Ambiguous Sentence Structure

We can use difficult English examples to illustrate how language can become ambiguous or difficult to parse.

Example:

“All the money he had had had had no effect on him.”

Correct interpretation requires identifying the grammatical role of repeated occurrences of the word “had.”

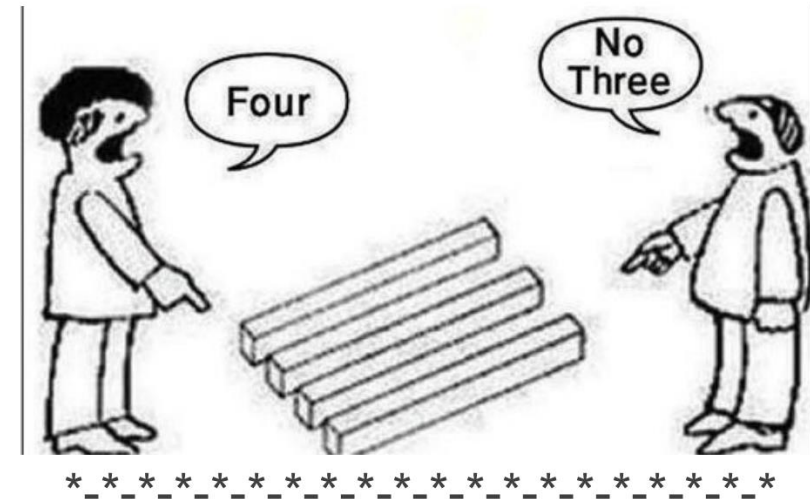
A second example demonstrates how long sequences of modifiers and participles can make syntactic interpretation extremely difficult.

Ambiguity Caused by Word Segmentation

We can introduce this concept through the expression “crashed blossoms,” referring to a phrase whose interpretation can change when the reader assigns different boundaries or meanings.

The same sequence of words can therefore trigger different interpretations depending on how the language is parsed.

For NLP systems, this illustrates why tokenization, segmentation, syntactic parsing, and contextual interpretation are closely related.



Ambiguity in Headlines

News headlines often compress information by omitting function words and using highly condensed syntax.

This can create interpretations that differ from the intended meaning.

We can give Arabic headline as an example in which the syntactic grouping leads the reader toward an unintended interpretation.

NLP systems must therefore recover implicit relationships and use context to infer the intended meaning.



Lexical and Syntactic Ambiguity in English

Take the following pair:

“A woman, without her man, is nothing.”

“A woman: without her, man is nothing.”

Changing punctuation changes the syntactic structure and consequently the interpretation.

This illustrates why punctuation and syntactic structure can be essential signals for language understanding.

Ambiguity in Sentence Parsing

The sentence “Federal raises interest rates 0.5%” can be segmented in several ways.

Possible groupings shown in the source include:

- (Federal) (raises) (interest rates) (0.5%)
- (Federal raises) (interest) (rates 0.5%)
- (Federal raises interest) (rates) (0.5%)

The example demonstrates that a parser must determine the correct syntactic structure before assigning meaning.

Even in the Holy Qur'an, there are **pausing signs intended to prevent an incorrect understanding:**

In Surah Al-An'am:

“إِنَّمَا يَسْتَجِيبُ الَّذِينَ يَسْمَعُونَ وَالْمَوْتَىٰ يَبْعَثُهُمُ اللَّهُ ثُمَّ إِلَيْهِ يُرْجَعُونَ”

- In Surah Al-Ma'idah, the verse can be understood in **two different ways depending on where the pause is made:**

•

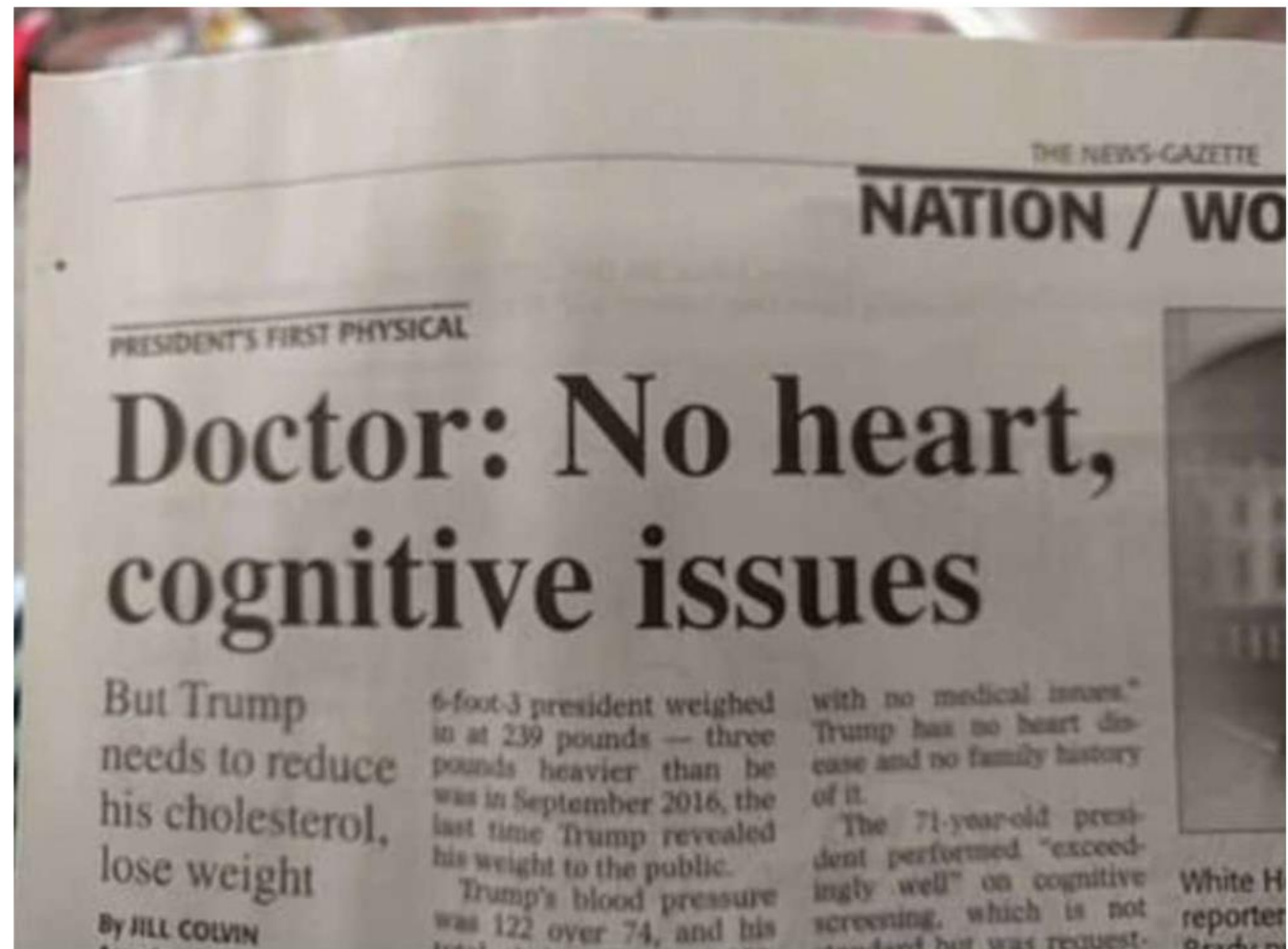
“قَالَ فَإِنَّهَا مُحَرَّمَةٌ عَلَيْهِمْ أَرْبَعِينَ سَنَةً يَتِيَهُونَ فِي الْأَرْضِ فَلَا تَأْسَ عَلَى الْقَوْمِ الْفَاسِقِينَ”

Context, Punctuation, and Interpretation

punctuation and stopping markers can help prevent incorrect interpretations.

This is particularly relevant to languages in which segmentation and interpretation may depend strongly on orthography, punctuation, and contextual cues.

Ambiguity in Real-World English



Headline Ambiguity and Compressed Language



www.bbc.co.uk/news/

Girl found alive in France murders car

A girl has been found hiding inside a British-registered car that was discovered eight hours earlier, French police say.

▶ Shock at 'brutality'

Teacher Strikes Idle Kids

Red Tape Holds Up New Bridges

Hospitals Are Sued by 7 Foot Doctors

Juvenile Court to Try Shooting Defendant

Local High School Dropouts Cut in Half

The “Crashed Blossoms” Example

The terminology is traced in the source to the headline:

“Violinist linked to JAL crash blossoms.”

The intended meaning concerns a violinist associated with the JAL crash; however, the wording can initially be interpreted as though “crash” and “blossoms” form a different semantic relation.

The source uses “crashed blossoms” / “crash blossoms” as a label for this kind of headline ambiguity.

Challenges in Arabic NLP

Large dialectal variation, even within the same language.

Informal language used in many conversations.

Incorrect or ambiguous sentence segmentation.

Idiomatic and figurative expressions.

Newly coined or rapidly evolving expressions.

Sarcastic uses of words and expressions.

Words with multiple meanings and uses.

Dependence on historical and world knowledge.

Difficulty identifying named entities and domain-specific expressions.

Examples of Arabic-Language Challenges

Dialectal variation: a word such as “شريحة” may have different meanings or uses in Morocco, Egypt, and the Gulf.

Informal expressions may differ substantially from formal written language.

Figurative expressions such as “ريح شوية” or “طير من هنا” cannot always be interpreted literally.

New expressions such as “فرمته” and “رنله” illustrate rapidly evolving colloquial vocabulary.

Sarcasm and context-dependent expressions can reverse or alter the apparent meaning.

A word such as “يا سلام” may express different meanings depending on context.

Historical and world knowledge can also be necessary to interpret statements correctly.

A Broader View of NLP Challenges

Dan Jurafsky



Why else is natural language understanding difficult?

non-standard English

Great job @justinbieber! Were SOO PROUD of what youve accomplished! U taught us 2 #neversaynever & you yourself should never give up either♥

segmentation issues

the New York-New Haven Railroad
the New York-New Haven Railroad

idioms

dark horse
get cold feet
lose face
throw in the towel

neologisms

unfriend
Retweet
bromance

world knowledge

Mary and Sue are sisters.
Mary and Sue are mothers.

tricky entity names

Where is *A Bug's Life* playing ...
Let It Be was recorded ...
... a mutation on the *for* gene ...

But that's what makes it fun!

Key NLP Challenges : Summary

Ambiguity

Multiple valid interpretations of the same surface form.

Segmentation

Determining meaningful boundaries between words, phrases, or clauses.

Syntax

Recovering the grammatical structure of a sentence.

Semantics

Determining the intended meaning of words and sentences.

Context & Knowledge

Using discourse context and world knowledge to resolve meaning.

Language Variation

Handling dialects, informal language, neologisms, sarcasm, and idioms.

Key Takeaway

The central difficulty in NLP is not simply processing text; it is recovering the intended meaning behind language.

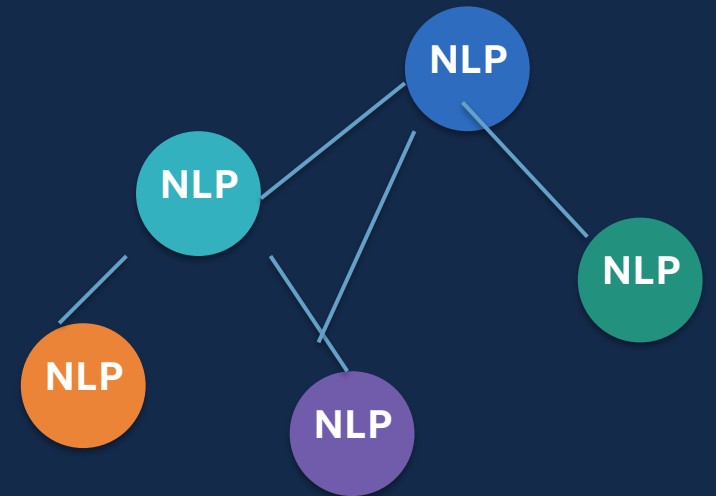
Ambiguity may be lexical, syntactic, semantic, contextual, or pragmatic.

Real-world NLP must also cope with dialects, informal language, idioms, neologisms, sarcasm, named entities, and incomplete world knowledge.

The examples in this chapter demonstrate why robust language understanding requires more than word-level processing.

Applications of NLP

From unstructured text to actionable information



Where NLP Creates Value

IR

Information Retrieval

Find relevant documents, passages, or facts.

IE

Information Extraction

Convert text into structured entities, events, and attributes.

SA

Sentiment Analysis

Estimate attitudes, opinions, and emotional polarity.

QA

Question Answering

Return answers to natural-language questions.

MT

Machine Translation

Transform content from one language into another.

CA

Conversational AI

Enable text- and voice-based interaction.

From an Email to a Calendar Event

NLP can identify actionable information embedded in an ordinary message.

✉ Email

Hi Dan, we've now scheduled the curriculum meeting.

It will be in Gates 159 tomorrow from 10:00–11:30.

— Chris

Extracted information

EVENT Curriculum meeting

LOCATION Gates 159

DATE Tomorrow

TIME 10:00–11:30

Named entities + temporal expressions + relations

Opinion Mining from Product Reviews

Instead of treating an entire review as a single label, NLP can separate the product aspects being discussed.

Customer review

“The camera is small and light,
easy to carry, but the body feels
flimsy.”

Aspect-level interpretation

Size & weight

POSITIVE

Portability

POSITIVE

Build quality

NEGATIVE

Output: structured opinion data that can be aggregated across many reviews.

Core NLP Application Families



A Broader Set of NLP Use Cases



Translation

Language conversion



Fake News Detection

Classify questionable content



Email Classification

Route and label messages



Disease Prediction

Extract signals from language



Error Detection

Identify anomalies



IVR Applications

Speech-driven interaction



Sentiment Analysis

Estimate expressed attitude



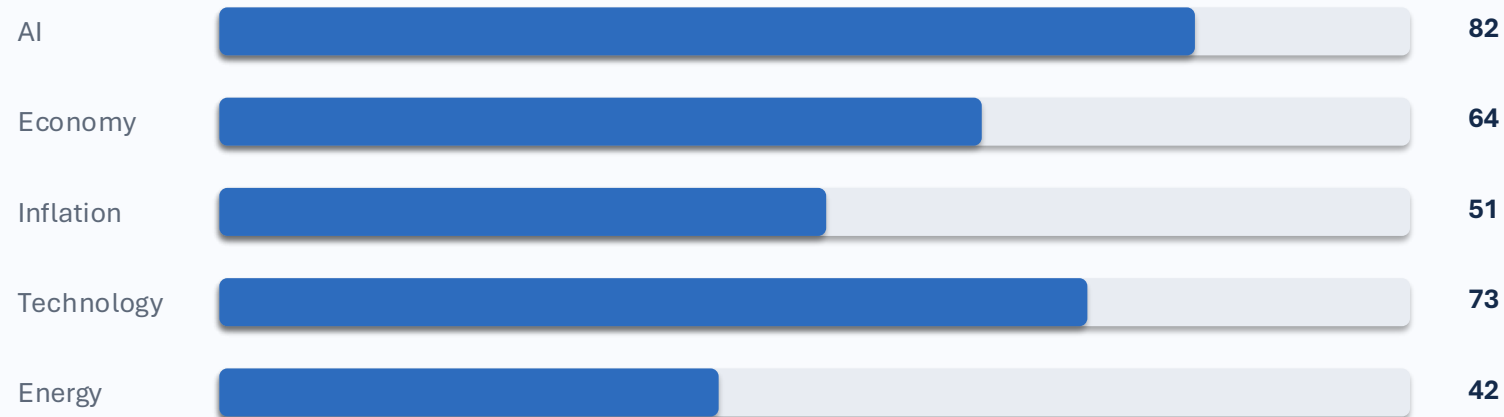
Personal Voice Assistants

Conversational commands

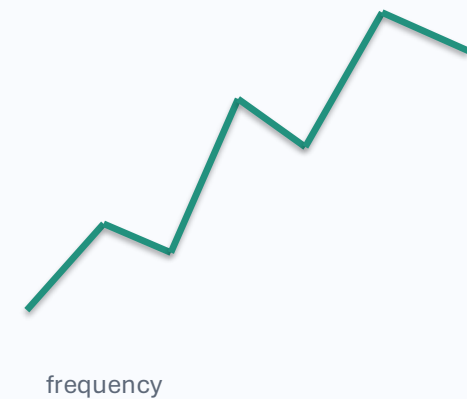
Mining Language from the Web

Some NLP systems automatically collect words and phrases from selected websites, then analyze their frequency and usage over time.

Text analytics dashboard



Usage over time



<http://sentedx.com/financial-analysis/>

Political Text Analytics

political-language analysis could be included an example of large-scale text analytics.



The objective is descriptive analysis of language data—not a political conclusion.

<http://sentdex.com/political-analysis/>

Geographical Analysis of Tweets

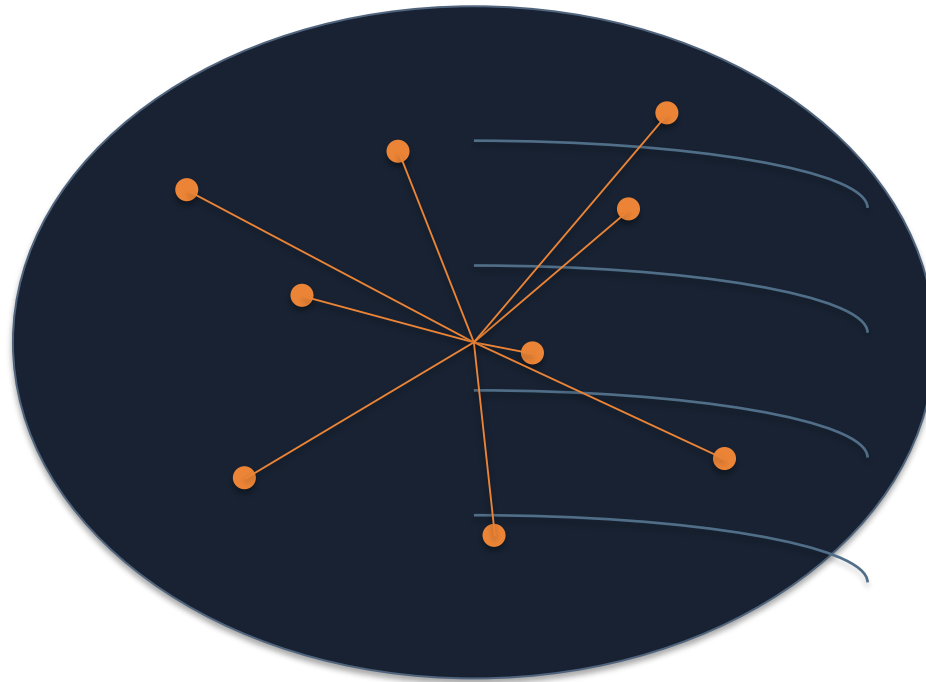
When text is associated with geographic information, NLP can be combined with geospatial analysis to reveal where linguistic patterns occur.

Tweets / posts

Location

Language patterns

**Geospatial
visualization**



<http://sentedex.com/geographical-analysis/>

NLP Applications by Difficulty

Mostly solved

Spam detection

ex: let's go to Agral
buy printer

POS tagging

Colorless green ideas sleep furiously.
ADJ ADJ. NOUN. VERB ADV

Named-entity recognition (NER)

Einstein met with UN officials in Princeton
PERSON ORG LOC

Making good progress

Sentiment analysis

Coreference resolution

Ex: Carter told Moubarak he should'nt
run again.

Word-sense disambiguation

Ex: I need a new batterie for my mouse.

Parsing

Machine translation

Information extraction

Still a little hard

Question answering

Paraphrase

Ex: XYZ acquired ABC yesterday
ABC has been taken over by XYZ

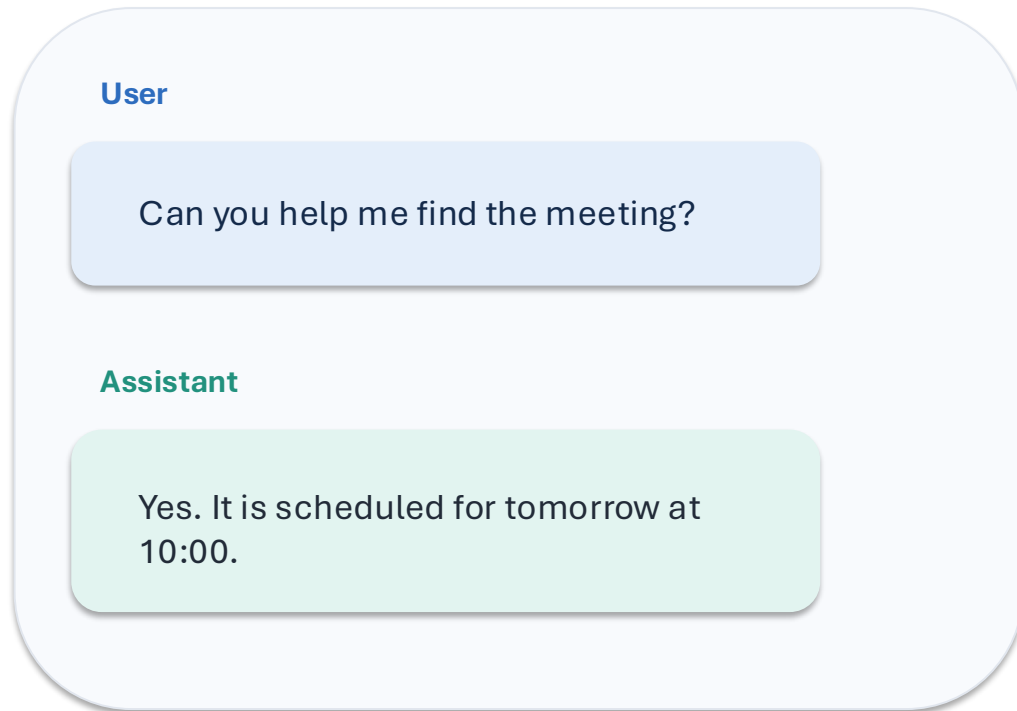
Summarization

Open-ended dialogue

Difficulty is task-dependent and changes as models, data, and evaluation methods evolve.

Chatbots and Conversational AI

Chatbots both text-based and voice-based as an important application area for NLP.



Typical NLP pipeline



Main Conclusions



NLP turns language into usable information.

Extraction, retrieval, classification, and analysis are core application patterns.



Applications span multiple modalities. Text, web content, social media, email, and voice can all become NLP inputs.



The same language technology supports different objectives.

Search, sentiment analysis, question answering, translation, and dialogue require different pipelines.



Task difficulty varies.

Some tasks are mature, while open-ended language understanding remains challenging.